

Federated Learning for Generalization, Robustness, Fairness: A Survey and Benchmark

Wenke Huang[✉], Mang Ye[✉], *Senior Member, IEEE*, Zekun Shi[✉], Guancheng Wan, He Li[✉],
Bo Du[✉], *Senior Member, IEEE*, and Qiang Yang[✉], *Fellow, IEEE*

Abstract—Federated learning has emerged as a promising paradigm for privacy-preserving collaboration among different parties. Recently, with the popularity of federated learning, an influx of approaches have delivered towards different realistic challenges. In this survey, we provide a systematic overview of the important and recent developments of research on federated learning. First, we introduce the study history and terminology definition of this area. Then, we comprehensively review three basic lines of research: generalization, robustness, and fairness, by introducing their respective background concepts, task settings, and main challenges. We also offer a detailed overview of representative literature on both methods and datasets. We further benchmark the reviewed methods on several well-known datasets. Finally, we point out several open issues in this field and suggest opportunities for further research.

Index Terms—Fairness, federated learning, generalization, robustness.

I. INTRODUCTION

CURRENT development of deep learning has caused significant changes in numerous research fields, and had profound impacts on every nook and cranny of societal and industrial sectors, including computer vision [1], [2], [3], [4], natural language processing [5], [6], multi-modal learning [7], [8], [9], medical analysis [10], graph learning [11], [12], and more. However, the success of deep learning heavily relies on large-scale data and there has been increasing awareness in the public and scientific communities for data privacy. Specifically, in the real world, data is commonly distributed among different entities (e.g., edge devices and companies). Due to the increasing emphasis on data sensitivity, strict legislations [13], [14], [15]

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Wenke Huang, Mang Ye, Zekun Shi, Guancheng Wan, He Li, and Bo Du are with the School of Computer Science, Wuhan University, Wuhan 430072, China (e-mail: wenkehuang@whu.edu.cn; yemang@whu.edu.cn).

Qiang Yang is with the Department of Computer Science and Engineering, Hong Kong University of Science and Technology, Hong Kong, China (e-mail: qyang@cse.ust.hk).

We also provide a public website to continuously track developments in this fast advancing field: <https://github.com/WenkeHuang/MarsFL>.

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have been proposed to govern data collection and utilization. Thus, the traditional centralized training paradigm, which requires to aggregate data, fails to deploy in realistic scenarios. Driven by such realistic challenges, federated learning (FL) [16], [17], [18] has emerged as a popular research field because it can train a global model for different participants without centralizing data owned by the distributed parties. Roughly speaking, the classical federated paradigm can be abstractly divided into the following two steps: server-side collaboration and client-side optimization. The former could be regarded as that a central server aggregates parameters from participants and then distributes the global model (averaged parameters) back. The latter represents that the client optimizes the distributed model on the local private data. Therefore, FL achieves the privacy-preserving collaboration to learn a shared model without data consolidation. Despite great advancements in federated learning [19], [20], [21], [22], [23], current federation has three major challenges as:

- **Generalization:** The distributed data is normally collected from different sources with diverse preferences and naturally brings the non-independent and identically distributed (Non-IID) characteristics [24], [25], [26], [27], [28], [29], [30]. Owing to this distribution discrepancy, there exists two major distribution shift types.

- i) **Cross-Client Shift:** Distributed data from different clients is inevitable to appear a different underlying distribution, incurring large heterogeneity among client data. Under this circumstance, each client optimizes toward its distinct local empirical minimum, resulting in divergent local optimization directions [31], [32], [33], [34], [35]. Thus, the federation presents slow convergence and unsatisfying performance.

- ii) **Out-Client Shift:** The federated system only incorporates knowledge from participating clients and optimizes toward fitting online client distribution. Therefore, it would be vulnerable under the Out-Client Shift [36], [37], [38], [39], [40]. Specifically, when deploying the federated model to outer client distribution (i.e., unseen testing domains), the federation fails to adapt to the target distribution and inevitable performance degradation is commonly observed. All in all, achieving a generalizable federated model acts as a foundational requirement to boost the convergence speed and performance accuracy for the real-world data distribution.

- **Robustness:** As a privacy-aware collaborative paradigm, ensuring federated robustness plays a crucial role in guaranteeing federated effectiveness. In practice, due to its distributed nature,

federated learning fails to ensure the client trustworthiness and is highly vulnerable to different malicious behaviors. A small set of malicious clients can easily manipulate the training process by uploading poisoned local models to the server

i) Byzantine Attack: malicious clients endeavor to falsify real network updates to attack the federated convergence and performance, [41], [42], [43], [44]. Existing attacks mainly focus on poisoning client local training data, i.e., Data-Based Byzantine Attack [45], [46] and directly manipulating the model parameters, Model-Based Byzantine Attack [47], [48], [49]

i) Backdoor Attack: evils inject the backdoor trigger to induce the global model to learn misleading information, so as to predict attacker-chosen labels for adversarial examples while maintaining stable performance on the main task [50], [51], [52], [53]. Furthermore, backdoor attacks utilize the distributed paradigm to consider more stealthy trigger patterns via decomposing the global trigger pattern into separate local patterns to easily dodge various defensive mechanisms while successfully launching distributed backdoor attacks [54], [55].

• **Fairness:** Federated learning functions as a collaborative paradigm. The crucial cooperation pre-requirement is to satisfy the multi-party interest allocation for its sustained development viability [56], [57], [58]. Within this context, two primary interest conflicts groups emerge out.

i) Collaboration Fairness: In the absence of equitable credit assignment and just reward allocation, clients lack the incentive to actively participate in the federated collaboration [59], [60], [61].

ii) Performance Fairness: The federated model may result in biased predictions, thereby yielding sub-optimal performance for clients characterized by limited data scale or distinct data distributions from the rest [62], [63], [64]. Consequently, the maintenance of both collaboration fairness and performance fairness significantly determines whether federated learning can function in a robust and enduring manner.

These shortcomings severely hinder the adoption of federated learning in realistic applications, such as medical diagnosis [65], autonomous driving [66], and market finance analysis [67], which lead to the increasing concern on the contemporary federated learning techniques.

A. Prior Surveys

As FL research has become a prominent research field in recent years, a large amount federated survey papers have emerged. Pioneering works [69], [71], [74], [76], [86] basically focus on the conceptual framework and macro guidance and neglect in-depth exploration of specific challenges and problems. The majority focuses on the data heterogeneity problem [68], [73], [77], [79], [80], [83], [85], [87] from different angles. For example [68], [79], [85] tackle the non-IID issue in FL and provide detailed discussion on different specific data heterogeneity forms with respective federated solutions. A few attempts respectively investigate the robustness [57], [69], [71], [78], [84] and fairness [58], [76] issues. Some surveys discuss specific real-world applications such as medical contexts [88] and smart healthcare [65] but lack the universality for different scenarios. For instance, [65] concentrates on the federated healthcare topic,

TABLE I
OVERVIEW OF RELATED SURVEYS

Work	Generalization	Robustness	Fairness	Benchmark
[arXiv'18] [68]	Cross-Client Shift			
[TIST'19] [69]	Cross-Client Shift			
[WS4'20] [70]	Cross-Client Shift			
[SPM'20] [71]	Cross-Client Shift	Byzantine		
[arXiv'20] [72]		✓		
[NC'21] [73]	Cross-Client Shift			
[FTML'21] [74]	Cross-Client Shift	✓	✓	
[FGCS'21] [75]	Cross-Client Shift	✓		
[TKDE'21] [76]	Cross-Client Shift	✓	Performance	
[arXiv'21] [77]	Cross-Client Shift	✓		
[TrustCom'22] [78]		Byzantine		✓
[TKDE'22] [79]	Cross-Client Shift			✓
[CSUR'22] [65]	Cross-Client Shift			
[arXiv'22] [80]	Cross-Client Shift			✓
[TNNLS'22] [81]		✓		
[FGCS'22] [82]	Cross-Client Shift			
[arXiv'23] [57]			✓	
[arXiv'23] [83]	Out-Client Shift			
[ariv'23] [84]	Cross-Client Shift			✓
[TNNLS'23] [58]			✓	
[CSUR'23] [85]	Cross-Client Shift			
Ours	✓	✓	✓	✓

See details in § 1.1.

which mainly discusses scalable smart healthcare networks and applications, including healthcare management, remote health monitoring, and so on. However, over-focusing on specific areas weakens their universality in other federated scenarios. Importantly, although there are few surveys such as [78], [79], [80], [84] with empirical experiments analysis, their framework appears a strong entanglement with the specific problems and thus lacks the high flexibility towards realistic federated problems. Besides, we provide fruitful SOTA methods implementations, which largely benefit relative researchers. All in all, with the rapid advance of this field, **Generalization**, **Robustness**, and **Fairness** have been crucial aspects in federated learning. Generalization ensures federated performance on heterogeneous distribution. Robustness guarantees the federation to defend against malicious attacks. Fairness provides reasonable interest allocation to maintain sustainable multiple-party collaboration. Although there is a huge body of new literature, most existing surveys focus on the **narrow view** with **fragmented results**. In contrast, we argue that these three pieces interact with each other to jointly enhance the practical federation deployment and this is the first work to **simultaneously investigate** the related research development and **uniformly benchmark** multi-view experimental analysis on the Generalization, Robustness, and Fairness realms. We provide a comprehensive comparison in Table I.

B. Contribution

In this paper, we introduce a comprehensive survey on federated learning and mainly focus on generalization, robustness, and fairness. Compared with existing surveys, this paper makes the following contributions:

- We delve into an in-depth exploration of federated learning and provide the first state-of-the-art and symmetric survey on the generalization for heterogeneous distribution,

TABLE II
SUMMARY OF ESSENTIAL CHARACTERISTICS FOR REVIEWED METHODS IN
CROSS CALIBRATION (SECTION III-B)

Methods	Venue	Highlight
<i>Global Neural Network</i>		
Limitation: Linearly increase the local computation cost		
FedProx[27]	[MLSys'20]	Parameter ℓ_2 normalization
SCAFFOLD[164]	[ICML'20]	Control variate
MOON[31]	[CVPR'21]	Feature-level contrastive learning
FedNTD[169]	[NeurIPS'22]	Decoupled distillation
FedSeg[90]	[CVPR'23]	Pixel-level contrastive learning
<i>Global Statistic Information</i>		
Limitation: Reliance on data richness for ideal initialization		
FedProc[174]	[arXiv'21]	Prototypical contrast
HarmoFL [175]	[AAAI'22]	Normalized amplitude
FedFA [176]	[ICLR'23]	Gaussian augmentation
FPL[111]	[CVPR'23]	Cluster and unbiased prototypes
<i>Extra Network Architecture</i>		
Limitation: Compatibility difficulty and communication burden		
FedMLB[177]	[ICML'22]	Multiple auxiliary branches
FedCGAN[178]	[IJCAI'22]	Conditional GAN
ADCOL[179]	[ICML'23]	Representation generator
DaFKD[180]	[CVPR'23]	Discriminator module
<i>Self-Driven Regularization</i>		
Limitation: Forgetting with unstable hyper-parameter		
FedRS[181]	[KDD'21]	Restrict Softmax function
FedAlign [182]	[CVPR'22]	Final block Lipschitz constant
FedSAM [35]	[ICML'22]	Sharpness aware minimization
FedLC [97]	[ICML'22]	Calibrate logit via class probability
FedDecor[183]	[ICLR'23]	Dimensional decorrelation
<i>Federated Data Sharing</i>		
Limitation: Pre-defined relevant data is not always available		
DC-Adam [184]	[CS'21]	Pre-shared data for warm-up strategy
FEDAUX [185]	[TNNLS'21]	Pretrain and distill on auxiliary data
<i>Federated Data Enhancement</i>		
Limitation: Limited diversity and potential privacy leakage		
FedMix[186]	[ICLR'21]	Close global Mixup via averaged data
FEDGEN[187]	[ICML'21]	Data generator ensemble learning
<i>Federated Data Selection</i>		
Limitation: Data and client-level unfairness		
FedACS[188]	[IJWQS'21]	Filter out poisoned data via clustering
Safe[189]	[TII'22]	Thompson sampling low skew clients
<i>Server Aggregation Reweighting</i>		
Limitation: Require qualified datasets and element evaluation		
FEDBE[190]	[ICLR'21]	Bayesian ensemble inference
Elastic [191]	[CVPR'23]	Interpolate parameter via sensitivity
<i>Server Adaptive Optimization</i>		
Limitation: Demand proxy dataset and elaborate joint-goal		
FedMD [192]	[NeurIPS'19]	Additional classifier for data distillation
FedDF [193]	[NeurIPS'20]	Ensemble distillation on related data
FedGKT[194]	[NeurIPS'20]	Server-side group knowledge transfer
FedOPT[195]	[ICLR'21]	Federated adaptive server optimizer
FCCL [196]	[CVPR'22]	Cross-Correlation on unrelated data

robustness for malicious attacks, and fairness for multiple-party interest in federated learning, including hundreds of papers in this fast-growing field.

- We select influential works published in prestigious journals and conferences and classify the existing federated methods, based on different task settings: Cross-Client Shift and Out-Client Shift in Generalizable Federated Learning, Byzantine Attack and Backdoor Attack in Robust Federated Learning, Reward Conflict and Prediction Biases in Fair Federated Learning. Except for the

TABLE III
SUMMARY OF ESSENTIAL CHARACTERISTICS FOR REVIEWED SOLUTIONS FOR
UNKNOWN GENERALIZATION (SECTION III-C)

Methods	Venue	Highlight	Limitation
<i>Federated Domain Adaptation § 3.3.1</i>			
FADA [37]	[ICLR'20]	Dynamic adversarial alignment	◦: GAN [200]
COPA [242]	[ICCV'21]	Shared extractor & group heads	◦: IBN [243]
AEGR [244]	[ICME'23]	Pseudo-label refinement	*: PGD [245]
<i>Federated Domain Generalization § 3.3.2</i>			
FedDG [38]	[CVPR'21]	Sharing amplitude spectrum	*: Amplitude
CCST [100]	[WACV'23]	Cross-client style transfer	*: Style info
CSAC [246]	[TKDE'23]	Layer semantic aggregation	◦: Attention

* and ◦ respectively represent the potential privacy leakage and network architecture modification.

taxonomies, in-depth analysis about the pros and cons of these methods is also provided.

- We conduct a plentiful benchmark analysis on various federated scenarios with different solutions. Alongside a set of evaluation metrics: generalizable performance, defensive degree, and fair level, we comprehensively investigate the methods effectiveness.
- We discuss future research directions, which will assist the community to rethink and improve their current designs toward federated learning in real practicality. Meanwhile, in order to promote the development of this field.

The remainder of this paper is organized as follows. Fig. 1 shows the structure of this survey. Section II gives some brief introduction on the history, terminology, and problem definition. We review representative papers on federated algorithms from three aspects: Generalizable Federated Learning in Section III, Robust Federated Learning in Section IV, and Fair Federated Learning in Section V, respectively. Section VI conducts performance evaluation and analysis. While Section VII-A raises open questions and potential directions. Finally, we make concluding remarks in Section VII-B.

II. BACKGROUND

A. History and Terminology

Federated learning typically aims to enable systems or devices to collaboratively construct a global model while maintaining all private data locally, without the need for explicit data exchange among clients. This concept was introduced in milestone work such as [16], [17], [89]. Based on distribution characteristics, existing federated learning can be categorized into the following types [73], [86]: Horizontal Federated Learning (HFL): Clients own the same feature space while having different sample spaces, and collaboratively optimize a shared global model [89], [90]. Vertical Federated Learning (VFL): Each device has a dataset with distinct feature characteristics derived from the same sample identification space [91], [92]. Thus, Vertical Federated Learning can be seen as performing feature-level privacy-preserving computations. Federated Transfer Learning (FTL): When both sample spaces and feature spaces are different among clients, FTL basically leverages the transfer learning technique to achieve knowledge communication [93], [94], [94].

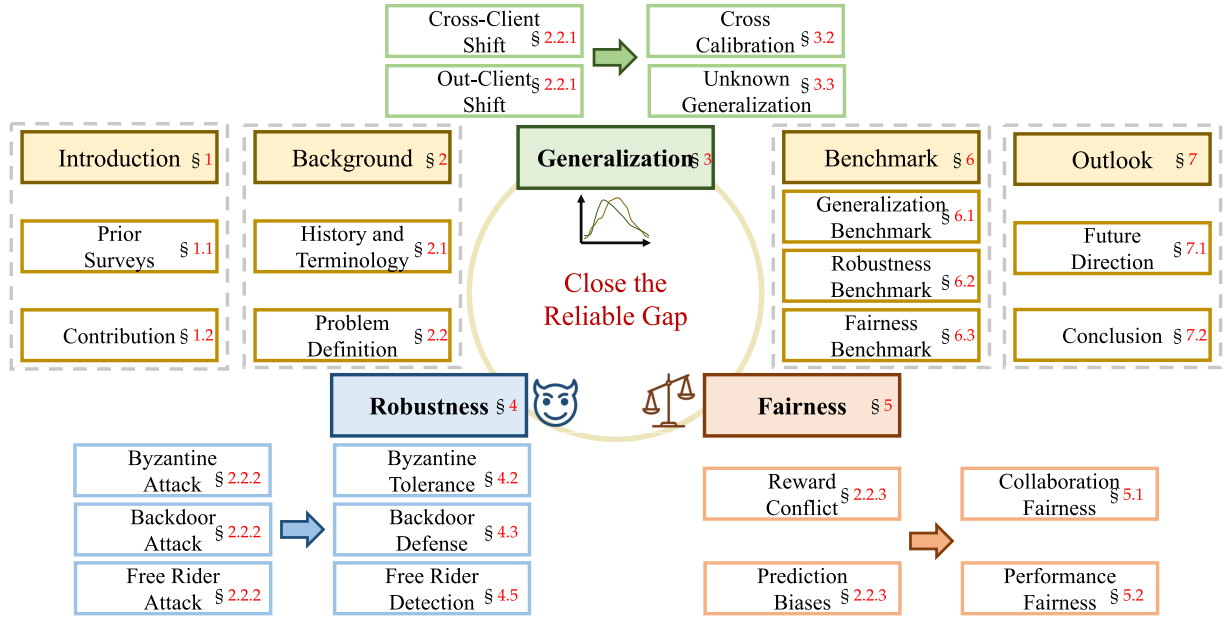


Fig. 1. Overview of the survey. Best viewed in color.

In this paper, we focus on three crucial characteristics in federated learning and we conduct the uniform benchmark evaluation on the Horizontal Federated Learning¹ (HFL) scenario. *I) Generalizable Federated Learning (GFL)*: As a slang goes that when you hear hoofbeats, think of horses, not zebras. During federated learning, the decentralized data poses non-independent and identically distribution (called data heterogeneity) [37], [83], [95], [96]. Then, there naturally exist two types of shift. First, Cross-Client Shift denotes the difference between empirical risk loss among participating clients [97], [98], [99]. Second, Out-Client Shift means the difference in expected risk between participating and non-participating clients [37], [100], [101]. *II) Robust Federated Learning (RFL)*: Recent works show that standard federated learning is vulnerable to malicious attacks. One stream, Byzantine Attack, investigates polluting the local data or the uploaded model to jeopardize the discrimination and convergence of the federated model [41], [102], [103]. The other group, Backdoor Attack aims to induce the local model to incorporate misdefined information and activate the backdoor trigger to achieve malicious targeted prediction [53], [104], [105], [106]. *III) Fair Federated Learning (FFL)*: Fairness in federated learning mainly encompasses two aspects [57], [81], [107]. During the training stage, clients need to optimize the distributed model on their local data. The local updating process incurs computation cost to train the model and the training effectiveness largely depends on the data quality [59], [108]. Consequently, the computation cost and data value should be carefully considered to ensure the Collaboration Fairness. As the saying goes *No pain, no gain*, federated learning needs to estimate reward to incentivize the client to contribute to the federated system. Besides, given the distributed data distribution, fitting different client data poses varying difficulty degrees. Naturally, during the inference period, the shared model would manifest the biased performance of minority or marginalized groups [62],

[109], [110]. Thus, maintaining Performance Fairness becomes an essential objective to mitigate performance disparities.

B. Problem Definition

Following the standard federated learning setup [27], [31], [89], [97], [111], suppose there are M clients, (indexed by i) with respective private data D_i . $N_i = |D_i|$ means the private data scale for the i th client private dataset. We further assume each data sample (x, y) , where x is the input attribute and y is the label. The local data follows the distribution $\mathbb{P}_i(x, y)$. We denote the global model parameter as $w = f \circ g$, where f means the network backbone and g represents the uniform classifier. For the query sample $\xi = (x, y)$, The f maps x into the d dimensional feature vector $h = f(x) \in \mathbb{R}^d$. g maps feature h into logits output $z = g(h) \in \mathbb{R}^{|C|}$, where C denotes classification set. Formally, federated solutions generally seek to learn an *ideal* global model, w^* to minimize the weighted empirical loss among clients:

$$w^* = \min_w \sum_{i=1}^M \alpha_i F_i(w, D_i), \quad (1)$$

where α_i denotes the pre-allocated aggregation weight ($\sum_{i=1}^M \alpha_i = 1$) and is normally based on the data scale: $\alpha_i = \frac{N_i}{\sum_{i=1}^M N_i}$, or client scale: $\alpha_i = \frac{1}{M}$. $F_i(w, D_i)$ represents the empirical loss. We define the L_i as the client-specific loss function and further disassemble the federated process into the following three steps:

$$\begin{aligned} w_i &\leftarrow w && \text{Distribute} \\ \min_{w_i} \mathbb{E}_{\xi \in D_i} [L_i(w_i, \xi)] &&& \text{Optimize} \\ w &= \sum_{i=1}^M \alpha_i w_i && \text{Aggregate} \end{aligned} \quad (2)$$

¹We omit the word horizontal for brevity.

1) *Distributed Shift in FL*: The isolated data distributions are often non-independent and non-identically distributed (non-IID), reflecting local distribution differences among clients [27], [38], [112], [113], [114]. Preliminary, we define the i, j as the arbitrary client index among the participating clients M . And $\mathbb{P}_i(x, y)$ and $\mathbb{P}_j(x, y)$ respectively denotes the distribution for the client i and j . Furthermore, we utilize the $\mathbb{P}_o(x, y)$ to denote the out-of-client distribution. As for the distribution construction, the local distribution $\mathbb{P}_i(x, y)$ can be considered as $\mathbb{P}_i(x|y)\mathbb{P}_i(y)$. According to how the distribution $\mathbb{P}_i(x, y)$ is defined, distributed shift can be broadly categorized into two classes: Cross-Client Shift and Out-Client Shift.

- *Cross-Client Shift*: In the distributed manner, different clients normally hold the local data with different distributions. Thus, there exists the cross-client shift, which brings the distinct local empirical risk minima. Thus, the divergent client updating direction hinders the federated convergence speed and weakens federated effectiveness.

- i) *Label Skew*: The label distributions $\mathbb{P}_i(y)$ on the clients are distinct, and the conditional feature distribution $\mathbb{P}_i(x|y)$ is shared across clients. For instance, different hospitals specialize in distinct diseases and would hold inconsistent medical records ratio. Experiments normally leverage the Dirchlet sampling [115] to mimic the Label Skew. Furthermore, the Label Skew is formulated as the following form:

$$\begin{aligned} \mathbb{P}_i(y) &\neq \mathbb{P}_j(y) \\ \mathbb{P}_i(x|y) &= \mathbb{P}_j(x|y), \end{aligned} \quad (3)$$

where i and j denotes arbitrary clients in federation.

- ii) *Domain Skew*: As for the same label space, distinct feature distribution exists among different participants [28], [111], [116]. For the arbitrary clients i, j , conditional feature distribution $\mathbb{P}_i(x|y)$ and $\mathbb{P}_j(x|y)$ shows a certain difference. For example, cats may vary in coat colors and patterns in different body parts. We provide the mathematical definition in the following form:

$$\begin{aligned} \mathbb{P}_i(x|y) &\neq \mathbb{P}_j(x|y) \\ \mathbb{P}_i(y) &= \mathbb{P}_j(y). \end{aligned} \quad (4)$$

- *Out-Client Shift*: As shown in (1), the federated system fits and promotes performance in cross-client distribution. But when deploying the federated model to outer client distribution (i.e., unseen domains), denoted as o , nonparticipating client data exhibits the distributional shift compared to the participating client private data. Thus, out-client distribution $\mathbb{P}_o(x|y)$ shows distinct property with the cross-client distribution $\mathbb{P}_i(x|y)$. Then, the performance drop is commonly observed [36] because it fails to consider the underlying domain shift on the unseen distribution, an issue named Out-Client Shift. In our work, we conduct experiments based on the feature-level distribution skew and provide the following definition:

$$\begin{aligned} \mathbb{P}_i(x|y) &\neq \mathbb{P}_o(x|y) \\ \mathbb{P}_i(y) &= \mathbb{P}_o(y). \end{aligned} \quad (5)$$

- *Distinguish Out-Client Shift and Domain Generalization*: Domain generalization (DG) [117], [118], [119], [120], [121], [122], [123], [124], [125], [126], [127] focuses on enhancing the ability of models to generalize to an unseen test domain under the

centralized training paradigm, where a central server is feasible to access all source domain data. Therefore, DG seriously violates data privacy [14], [15]. However, Out-Client Shift operates within a federated system framework, allowing multiple participating domains to collaboratively learn a generalizable model for the out-client distribution. This paradigm is achieved without aggregating private data, embodying a decentralized solution.

- 2) *Malicious Behavior in Federated Learning*: Participants are uncontrollable in the distributed setting, some of them are malicious and aim to destroy the federated paradigm. We broadly divide adversarial behavior into three types, based on their destroying goal [128], [129]. As for Byzantine Attack [44], [102], [130], it is untargeted attack and the adversary objective is to hinder the model from achieving a near-optimal performance on the major task. With respect to Backdoor Attack [50], [51], [52], it could be assumed as the targeted attack, which ensures the optimized model presents differently on certain targeted sub-tasks while maintaining good overall performance on the primary task. Towards Free Rider Attack [131], [132], [133], [134], it focuses on plagiarizing other clients knowledge via crafting fake local updates to acquire the global shared model without really participating in the federation process.

- *Byzantine Attack*: Bateful client uploads the modified information to inhibit federated convergence and performance from the data-based and parameter-based angles.

- i) *Data-Based Byzantine Attack*: As for the data-based attacks [41], [42], malicious clients would intentionally pollute the local data to corrupt the learned model. We mainly consider the Symmetry Flipping (SymF) and Pair Flipping (PairF).

- ii) *Model-Based Byzantine Attack*: With respect to parameter-based [41], [48], [48], [53], [54], [104], [105], [135], evil manipulates updates before sharing with the server. We utilize the uploading gradient of the k participant as an example. For kind clients, they faithfully upload the gradient change as ∇_k . But malicious clients deliberately send distorted signals. We consider the following attacks types: Random Noise (RanN), Little Is Enough (LIE) [47], Fang [49], Min-Max (MiMa) [136], and Min-Sum (MiSu) [137]. We provide the detailed description in the appendix.

- *Backdoor Attack*: Compared with direct performance degradation, Backdoor Attack focuses on injecting a backdoor trigger pattern into the existing model while retaining the major task accuracy [104], [138], [139]. Specifically, we define the trigger pattern as the Φ and the modified sample as $\tilde{x} = (1 - \mathbf{m}) \odot x + \mathbf{m} \odot \Phi$. Thus, the original local optimization direction in (2) would be reformulated into:

$$\min_{w_i} \mathbb{E}_{(x, \tilde{x}, y) \in D_i} \left[L_i(w_i, x, y) + \varpi \underbrace{L(w_i, \tilde{x}, \tilde{y})}_{\text{Backdoor}} \right]. \quad (6)$$

The $\tilde{y}$ means the preset attack target predictions. The $\mathbf{m}$ means the trigger location mask. The hyper-parameter ϖ is utilized to control the trade-off between the original common empirical risk minimization and the backdoor injection loss function. Prior backdoor attacks are normally motivated by centralized backdoor attacks and assume that each malicious attacker independently trains their local models, without any collusion among them [53], [105]. Therefore, they utilize the same

backdoor trigger, the poisoned local models tend to appear with similar updates, largely deviated from the benign optimizing direction [140]. Recently, advanced attacks focus on the fixed-trigger backdoor paradigm [53], [54], [105], [141], [142], [143] and trigger-optimization backdoor solution [55], [144], [145], [146], [147], [148] to evade the secure detection strategies. To be precise, as for the fixed-trigger backdoor paradigm, DBA decomposes the global backdoor into separate ones, which makes DBA more stealthy from defenses. Both Neurotoxin [149] and BCLayer [142] consider selecting partial parameters elements that are less frequently updated to prevent backdoor erasing and backdoor-critical layers that are a small subset of layers and dominate the model vulnerabilities. FCBA [143] constructs an expanded combinatorial set of local triggers to amplify the resilience to the benign updates. Concerning the fixed-trigger backdoor paradigm, F3BA [146] fine-tunes the trigger pattern to accentuate the discrepancy between latent representations of unaltered and trigger-embedded samples. This method also directed gradients towards less critical model parameters. Concurrently, CerP [55] optimizes both the trigger and the model weights, employing l_2 regularization to reduce bias in the local model. IBA [148] collaboratively learns the optimal and visually stealthy backdoor trigger and then gradually implants the backdoor into the federated global model.

- **Free Rider Attack:** Another serious attack, Free Rider Attack, [128], [131] aims at disguising its participation to federated learning while ensuring that the iterative training process ultimately converges to the target. Thus, Free Rider Attack is a passive attack paradigm, which utilizes the distributed global model by the server but refuses to contribute valuable local information to the server [132], [150]. The motivation for free-rider could be concluded from two aspects. First, the adversary expects to save the computation and storage costs. Second, evils do not have qualified training data or are concerned about data privacy violations, so that local data are not available for model training. Existing Free Rider Attack solutions could be divided into two types. One is anonymous free-riders, who do not possess any private data and computation power. For instance, [131] proposes to upload the ordinary random weight and delta weight attack, which generates gradient updates by two global model subtractions. [151] introduces the scaled delta attack via the Ornstein-Uhlenbeck process to formalize the evolution of local and global training processes and analyze the geometrical relationship among client updates. Another type is selfish free-rider attacks, who own corresponding privacy datasets but are unwilling to devote themselves to collaborative training. Specifically, [132] presents the stochastic perturbations attack, which uploads crafted model weights by adding specific noises.

- **Static and Adaptive Attacks.** Traditional attack strategies often involve predefined and static behaviors, such as the use of Symmetry Flipping in Byzantine Attack [45], fixed trigger patterns in Backdoor Attack [43], [52], [54], and random weight in Free Rider Attack [131]. However, these methods tend to yield sub-optimal performance due to their lack of responsiveness to the global training dynamics. In contrast, adaptive attacks modify their strategies based on federated feedback or other observable data, underscoring the need for robust defense mechanisms.

For instance, A3FL [147] proposes to adversarially adapt the backdoor trigger to be persistent in global training dynamics. CerP [55] jointly control the poisoned model changes on each malicious attacker and implant the trigger pattern. Min-Max and Min-Sum [137] consider modifying the malicious client updates limited by the maximum and mean value distance with other benign clients. Scaled delta attack [151] analyzes the geometrical relationship of clients local model updates and formulates the local and global training processes via the Ornstein-Uhlenbeck process.

3) **Interest Conflict in Federated Learning:** Unlike the traditional centralized training paradigm, which optimizes on local data, federated learning involves multiple client collaboration. Thus, as a saying goes *No rules, no justice*. The federation needs to carefully deal with the client interest conflict to satisfy the expected performance improvement and reasonable benefit allocation [58], [152]. We mainly discuss several types of interest conflict:

- **Reward Conflict:** Existing federated literature is normally under the assumption that data owners have an inherent willingness to participate in the federated system [27], [89], [153]. However, during the federated process, the contributions of various clients are not uniform due to differences in local data value and computational costs [154]. Consequently, it becomes necessary to ensure that the client reward is proportionate to its contribution to the federation. Specifically, a participant contributing significantly should receive a higher reward than one with a lower contribution. Normally, existing research leverages the Game theory [155], which investigates the strategic interactions among rational agents. In our work, we utilize the Shapley Value [156], which is a marginal contribution-based scheme from cooperative game theory [157], [158] to evaluate the client contribution. To be precise, we consider the standard test dataset u for performance evaluation. We denote the online client set $S \subseteq M$ and D_S as the union dataset set. The overall accuracy on testing set u accuracy of federated model w , optimized on D_S , is defined as $A(w, D_S, u)$, abbreviated as A_S^u . The Shapley value for the client i , $\nu_i(w, D_S, D_i, u)$, abbreviated as ν_i , is formulated as the following form:

$$\nu_i = \frac{\rho}{|M|} \sum_{S \subseteq M \setminus \{i\}} \frac{A_{S \cup \{i\}}^u - A_S^u}{\binom{|M| - 1}{|S|}}, \quad (7)$$

where ρ means the constant parameter to rescale the shapely value. Via the (7), we assess the importance of client contribution via evaluating its marginal effect for the respective final model accuracy.

- **Prediction Biases:** Due to the data heterogeneity in the federated paradigm, it is possible that the federated performance would vary significantly with uneven performance on different distributions [62], [63]. For instance, under the Domain Skew (4), private data is normally derived from different domains with diverse distribution, e.g., MNIST [159] and SVHN [160] in the Digits scenario. Regrettably, solving the optimization in (1) does not inherently promote the shared global model to produce uniform predictive accuracy across multiple domains, referred to as Prediction Biases. Specifically, due to the fitting difficulty

and similarity correlation [161], the federated model would be biased towards specific domains and thus present inferior performance on those domains with hard or distinct distribution with others [110], [162]. We mathematically define the metric to evaluate the Prediction Biases degree as:

$$\zeta = \sigma\{\mathcal{A}^u\}_{u \in \mathcal{U}} \quad (8)$$

The σ means the standard deviation and $\mathcal{U}$ denotes the testing dataset collection. We assume that all clients join the federation and omit the A_M^T as A^T . The larger the ζ is, the more biased the predictive results are.

III. GENERALIZABLE FEDERATED LEARNING

A. Generalization Metrics

To achieve the generalization property during federation, existing methodologies for Cross-Client Shift and Out-Client Shift problems, are divided into two major streams: Cross Calibration (Section III-B) and Unknown Generalization (Section III-C).

- **Cross-Client Accuracy:** $\mathcal{A}^{\mathcal{U}}$: As for the Cross-Client Shift methods Section III-B, it mainly considers that the client distribution is sampled from the single domain distribution, Label Skew or multiple domain distributions, Domain Skew. Thus, when it comes to the evaluation stage, we abstract it as a test dataset collection with an uncertain scale, $\mathcal{U}$. We default assume that clients are sampled from several source domains and there is a testing dataset collection $\mathcal{U} = \{u\}$ and the unknown testing dataset O . We acquire the logits output ($z = w(x)$) from the global model on the query testing sample, $\xi = (x, y)$. We measure the standard Top-1 accuracy metric on each specific testing dataset $u \in \mathcal{U}$. We further utilize the average value to represent the performance in the following formulation:

$$\begin{aligned} \mathcal{A}^u &= \frac{\sum(\max(z) = y)}{|\mathcal{U}|}, \\ \mathcal{A}^{\mathcal{U}} &= \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \mathcal{A}^u. \end{aligned} \quad (9)$$

- **Out-Client Accuracy:** $\mathcal{A}^O$. Similarly, with respect to the Out-Client Shift solutions Section III-C, it focuses on evaluating the unknown distribution. Thus, we directly measure the Top-1 accuracy metric on the unseen domain distribution O as the following formulation:

$$\mathcal{A}^O = \frac{\sum(\max(z) = y)}{|O|}. \quad (10)$$

B. Cross Calibration

As for the Cross-Client Shift problem, each local distribution is highly heterogeneous with others, and the local objective of each party is inconsistent among clients. Thus, different client optimizes towards distinct local empirical minimization, bringing the divergent optimization direction. Related methods focus on calibrating the client optimization direction via three major angles. We categorize existing methods in Table II.

1) Client Regularization:

- **Global Neural Network.** The most intuitive method is to directly use a shared global model for local optimization guidance

because it literally aggregates clients knowledge and naturally acts as the global knowledge representative. Some works focus on directly regulating the local network based on the parameter stiff [24], [27], [163] and parameter variance [34], [164], [165], [166], [167]. A handful of methods utilize the global model output on the private data to penalize the local direction [31], [33], [90], [168], [169], [170], [171], [172], [173]. However, involving a global model in the local optimization process deeply enlarges the local computation cost and linearly increases with the parameter scale.

- **Global Statistic Information:** Although the global model represents the overall federated knowledge, it fails to provide more fine-grained information. A group of federated methods has developed to construct the class-wise representative, e.g. prototype [111], [174], [197], [198] and Gaussian distribution [32], [176]. Some approaches make use of spectrum [175] and feature maps [199]. However, it is vital to note that generating reliable global statistical knowledge necessitates data enrichment. Without this preliminary, initial signals may introduce bias and result in biased models.

- **Extra Network Architecture:** A plethora of works switch to utilizing additional network structures such as generative adversarial networks [178], [180], [187], [200], CLIP ([201]) and auxiliary global branch [168], [177], [194]. Nevertheless, they largely restrict the participating network architecture selection scope, which needs to be compatible with the extra network architecture and increase the communication cost.

- **Self-Driven Regularization:** A stream calibrates the client drift by enhancing the local learning generality in a self-driven manner. Generally, it adopts the self-distillation paradigm [182], [202] or modified Cross Entropy term [97], [181], [183], [203]. Although these methods get rid of the additional shared signals cost, they are normally hyper-parameter sensitive and presents fragile under serious data heterogeneous scenarios.

2) Client Augmentation:

- **Federated Data Sharing.** To alleviate the damage of heterogeneous client data, directly sharing relevant datasets acts as a naive and simple solution during federation. Current solutions normally leverage the labeled samples for the warm-up operation [68], [184] and unlabeled instances for distillation [204] and pretraining [185] strategies. However, in scenarios characterized by data scarcity, it may be infeasible to select suitable related datasets for sharing.

- **Federated Data Enhancement:** The client shift problem originates from the inconsistency between the ideal global and local client distribution. A surge of efforts supplements local data to mimic the ideal data distribution via the batch normalization layer [205], [206], MixUp [186], [207], [208], [209], [210], data generative generator [205], [211]. However, federated data enhancement largely depends on the local data diversity and the privacy leakage potential.

- **Federated Data Selection:** Another direction proposes dynamic data selection without impacting the data collection and client optimization. Generally speaking, this approach either employs local data filtering strategies [189], [212] to select meaningful data while rejecting potentially tainted samples, or it establishes client selection paradigms [188], [213], [214], [215],

[216], [217] that prioritize those with lower degrees of data heterogeneity. However, it is worth noting that while federated data selection can significantly boost convergence speed, it unavoidably introduces unfairness at both the data and client levels since it tends to favor the selection of mainstream data or clients while potentially ignoring other diversified samples.

3) Server Operation:

- *Server Aggregation Reweighting*: Compared to modifying local training paradigms for calibrating client shift, existing research has confirmed that default fixed parameter aggregation (α_i in (1)) also leads to slow convergence and accuracy drop because it fails to effectively calibrate client divergence [26]. Thus, a dynamic parameter aggregation framework is helpful to improve federated convergence and performance. Related solutions can be mainly divided into several types. Some techniques utilize different post-processing methods to mitigate the model drift, e.g., Bayesian non-parametric theory [190], [218], [219], [220], MCMC [221], optimal transport [222], gradient-based optimization [223], [224], and shapley-value [225]. Another big family is built upon network characteristics [191], [226], [227], [228]. However, they either require unlabeled auxiliary datasets in the server or involve element-wise parameter evaluation, which is time-consuming and laborious.

- *Server Adaptive Optimization*: Notably, adaptive optimization has proven to be an effective solution for tailoring models to specific tasks. One approach focuses on server-side fine-tuning by synthesizing relevant information to facilitate empirical risk minimization [194], [229], [230], [231], [232], [233] or conducting the knowledge distillation on public dataset for knowledge transferring [192], [193], [196], [234], [235], [236], [237], [238]. Another stream investigates the server-side optimizing objective, including gradient-biased server optimizer [195] and global local joint objectives [239], [240], [241]. Clearly, such implicit target-appearance modeling strategy hurts flexibility and adaptivity due to the requirement for the proxy dataset or elaborate optimization definition.

C. Unknown Generalization

Previous research has demonstrated that deep neural networks tend to empirically overfit training data, leading to overconfident predictions [247], [248]. This phenomenon can be problematic in real-world scenarios [249], where even minor variations, i.e. domain shift, in the characteristics of deployment examples can result in significant performance drop [250], [251]. With respect to the federated system, most existing works only focus on improving model performance among clients, while ignoring model generalizability onto unseen domains outside the federation [36], [37], [38], [252]. Existing efforts can be mainly divided into two groups based on the accessible stage of the unseen domain: Federated Domain Adaptation (FDA Section III-C1) and Federated Domain Generalization (FDG Section III-C2). We provide relative methods conclusion in Table III.

1) *Federated Domain Adaptation*: There has been an explosive interest in the Federated Domain Adaptation (FDA) paradigm. FDA normally assumes to incorporate unlabeled

data from target domains during the federated learning process. Closely related methods mainly divide into two factions.

- *Federated Domain Alignment*: The naive solution for handling domain shift with the target domain is to directly align and harmonize the multiple domain distributions. Relevant methods typically incorporate the contrastive learning [253], [254], knowledge distillation [255], [256], [257], adversarial learning [244], gradient matching [258], [259]. In essence, this stream helps to align the representations of different domains, reducing the domain shift and thus enhancing the model generalizable ability across domains. However, a significant challenge stems from the problem setup, wherein the target domain is used during optimization but may not be accessible in many real-world scenarios.

- *Federated Domain Disentanglement*: This line of methods focuses on separating knowledge into domain-invariant and domain-specific components. Specifically, [37], [41] explore the extension of adversarial adaptation techniques to the federated setting. [242] explicitly disentangle domain-invariant feature extractor and the ensemble multiple domain-specific classifiers. [168] utilize the Mixture of Experts theory [260], [261] to adjust the domain expert and domain generalist via gating mechanism. Nevertheless, it is important to note that these methods often rely on additional networks or the preservation of specialized modules to achieve the decoupling objective.

2) *Federated Domain Generalization*: Federated Domain Generalization investigates deploying the models trained on sites of heterogeneous distributions, directly generalizing to unknown target clients with domain shift. Several pioneering works attempt to incorporate domain generalization problems into federated learning and can be generally categorized into two classes:

- *Federated Invariant Optimization*: A major stream of methods exploits calibrating the local training objective to alleviate the domain shift influence. Closely related solutions normally leverage the shared information, e.g., amplitude spectrum [38], style distribution [100] and Iterative Naive Barycenter [101] to learn domain invariant representation. Obviously, sharing data could be regarded as a breach of privacy [100], [262] and leads to a significant communication burden. Besides, a cluster of methods harness the additional modules or replace original network architecture to enhance the network generality and alleviate the domain biased trend, such as GAN [200] in FedADG [263], [264], Adain [265] in CCST [100], and IBN [243] in COPA [242]. However, implementing these approaches can be complicated because of adapting to various structures.

- *Federated Invariant Aggregation*: Yet another complementary line of work tries to modify the aggregation strategy to enhance the domain invariance ability. Some take inspiration from performance fairness to achieve multi-domain performance fairness in order to mitigate the generalization gap via the empirical variance balance [266], agnostic distribution fusion [267]. A crowd of works delves into parameter aggregation from a fine-grained perspective, such as layer-wise semantic calibration [246].

IV. ROBUST FEDERATED LEARNING

A. Robustness Metrics

While federated learning has emerged as a promising solution for many realistic settings, it often assumes that all clients are trustworthy and provide reliable information for the federation. However, due to its distributed nature, FL is vulnerable to various malicious manipulations by rogue clients. Existing Robust Federated Learning strategies can be broadly categorized into two categories respectively towards Byzantine Attack and Backdoor Attack: Byzantine Tolerance (Section IV-B) and Backdoor Defense (Section IV-C). We provide the corresponding evaluation in the following.

- **Accuracy Decline Impact:** $\mathcal{I}$: The malicious attack goal for the Byzantine Attack Section IV-B is to hinder the federated performance compared with the accuracy of the global model (A^u), optimized in the benign setting without any attack. We denote the $\mathcal{A}_{Byz}^u$ as the performance under the Byzantine Attack. We define the attack impact, $\mathcal{I}$, to depict the accuracy drop as the following formulation:

$$\mathcal{I} = A^u - \mathcal{A}_{Byz}^u. \quad (11)$$

- **Attack Success Rate.** $\mathcal{R}^u$: For the Backdoor Attack Section IV-C, we estimate the effectiveness by calculating the proportion of triggered samples classified as target labels. As illustrated in (6), we construct the trigger testing sample as $(\tilde{x}, \tilde{y})$. Thus, $\mathcal{R}$ on the selected testing domain u is formed as:

$$\mathcal{R}^u = \frac{\sum (\arg \max(\tilde{z}) = \tilde{y})}{|\tilde{T}|}. \quad (12)$$

B. Byzantine Tolerance

To combat byzantine attackers, designing robust aggregation has become an effective paradigm. Existing solutions can be generally categorized into three classes:

- 1) **Distance Base Tolerance:** A group of robust aggregation strategy models normally compare the participating clients update differences and regard those significantly far from the overall direction as malicious clients and then exclude them from the aggregation process [103], [130], [137], [140], [272], [273], [274]. For example, Multi Krum [130] selects the candidate gradient that is the closest to its neighboring clients. Fools-Gold [140] leverages cosine similarity to identify malicious clients and allocate low weight. FABA [272] removes the outliers far from the mean value of the uploaded gradient. However, they basically rely on the data homogeneity hypothesis, where each client shares the independent and identically distribution data distribution. Therefore, these methods are not applicable under data heterogeneous federated learning.

- 2) **Statistics Distribution Tolerance:** A handful of robust aggregation schemes focus on constructing diverse statistical criteria to select the circumvent the evil clients [44], [268], [269], [275]. For instance, RFA [269] calculates the geometric median with an alternating minimization function, which which empirically appears rapid convergence and can be interpreted as a numerically stable version of the Weiszfeld algorithm. Notably, RFA gets rid of collecting the client individual contribution and is agnostic to the corruption level. Bulyan [44] cooperates trimmed median [268] to conduct a two-step

TABLE IV
SUMMARY OF ESSENTIAL CHARACTERISTICS FOR REVIEWED BYZANTINE TOLERANCE SOLUTIONS (SECTION IV-B)

Methods	Venue	Highlight
<i>Distance Base Tolerance § 4.2.1</i>		
Limitation: Fail in data heterogeneous federated learning		
Multi Krum [130]	[NeurIPS'17]	Average candidate gradient via Krum
FoolsGold [140]	[arXiv'18]	Filter sybils via contribution similarity
DnC [137]	[NDSS'21]	SVD based spectral for outliers detection
<i>Statistics Distribution Tolerance § 4.2.2</i>		
Limitation: Require relative mathematical assumption		
Trim Median [268]	[ICML'18]	Remove via coordinatewise trimmed mean
Bulyan [44]	[ICML'18]	Agree each coordinate by major vectors
RFA [269]	[TSP'22]	Geometric median and smoothed Weiszfeld
<i>Proxy Dataset Tolerance § 4.2.3</i>		
Limitation: Require qualified dataset and homogeneity assumption		
FLTrust [270]	[NDSS'21]	Allocate score via ReLU-clipped similarity
Sageflow [271]	[NeurIPS'21]	Entropy filtering and loss reweighting

meta-aggregation algorithm. Despite the certain advantages, they are also sensitive to the degree of data heterogeneity. They normally hypothesize that distributed data is constrained into a certain range. Besides, these methods are generally based on the statistics assumption, which presents a strong prior limitation for the realistic application.

- 3) **Proxy Dataset Tolerance:** One recent line of robust aggregation solutions [224], [270], [271], [276] leverages the proxy data to conduct the additional performance evaluation. The motivation behind this is that the proxy dataset normally shares a similar or constant semantic space with the private data and is assumed to be clean. Therefore, utilizing the proxy dataset for related evaluation modules is a reliable and convincing approach to effectively discriminate the benign and malicious ones. For example, Sageflow [271] proposes an entropy-based filter and reweights aggregation based on empirical loss. FLTrust [270] introduces ReLU-clipped cosine similarity and allocates high trust scores for those reliable clients. Notably, they depend on the auxiliary related data for examination, which hampers their practicability. Therefore, existing methods acquire strong assumptions for the data homogeneity or qualified related proxy dataset, as illustrated in Table IV.

C. Backdoor Defense

The potential backdoor attacks act as a serious threat to the federated learning system and have attracted a large number of interest in diverse defensive solutions to mitigate different backdoor attacks. Based on the distinct defensive kernel, the federated backdoor defenses can be classified into three major categories:

- 1) **Model Refinement Defense:** This type of efforts focuses on refining the aggregated global model to erase the possible backdoor attacks such as fine-tuning [277], logits distillation [192], [193], [196], [278], pruning, [279], and specification [280]. However, these strategies require a high-scale proxy dataset to tune the global model, which presents realistic deployment for the federated learning setting [224]. Besides, Model Refinement Defense fails to guarantee the backdoor erasing degree. Moreover, without the additional optimization regularization, it needs careful and complicated hyper-parameter

TABLE V
QUANTITATIVE LABEL SKEW RESULTS IN TERM OF $\mathcal{A}^d$, $\mathcal{A}^u$, AND $\mathcal{E}$ (13), ON CIFAR-10, CIFAR-100, MNIST AND FASHION-MNIST SCENARIOS

Methods	Cifar-10					Cifar-100					MNIST					Fashion-MNIST				
	1.0	0.5	0.3	0.1	$\mathcal{E}$	1.0	0.5	0.3	0.1	$\mathcal{E}$	1.0	0.5	0.3	0.1	$\mathcal{E}$	1.0	0.5	0.3	0.1	$\mathcal{E}$
FedAvg [89]	70.64	66.96	63.92	60.43	0.354	68.47	69.72	69.21	68.92	0.213	99.44	<u>99.37</u>	99.13	98.76	0.602	89.94	89.87	83.82	90.15	0.462
FedProx [27]	71.22	67.16	64.88	61.03	0.423	72.37	70.19	63.48	<u>67.4</u>	0.773	99.15	99.41	99.32	98.73	0.134	89.87	89.97	88.69	83.57	0.524
SCAFFOLD [164]	70.77	68.33	68.34	60.83	/	<u>71.91</u>	72.76	<u>69.82</u>	68.24	/	99.41	99.12	98.95	96.95	/	89.83	89.73	88.32	81.27	/
FedNova [112]	<u>70.94</u>	67.06	66.42	64.05	/	70.12	67.11	63.86	27.91	/	99.42	99.29	99.22	99.88	/	90.20	89.81	89.03	<u>84.39</u>	/
MOON [31]	69.73	<u>68.07</u>	<u>66.48</u>	61.71	0.063	71.47	69.51	69.09	65.53	0.412	99.51	99.36	99.17	98.02	0.324	90.52	90.11	<u>88.95</u>	82.92	0.614
FedRS [181]	70.14	66.036	63.89	59.47	0.184	69.81	68.53	67.32	67.16	0.637	99.34	99.33	<u>99.23</u>	<u>98.93</u>	0.333	90.01	89.40	88.47	77.54	0.579
FedDyn [163]	70.59	67.80	64.39	60.52	0.488	71.48	<u>71.25</u>	70.28	66.81	0.583	<u>99.48</u>	99.31	99.10	98.71	0.059	<u>90.24</u>	89.97	88.59	82.92	0.533
FedOPT [195]	70.44	66.70	65.95	<u>63.10</u>	/	69.40	68.52	67.57	67.26	/	99.32	99.11	98.92	98.13	/	90.06	89.65	88.79	83.41	/
FedProto [98]	69.75	65.05	56.45	48.74	0.319	70.07	70.83	68.32	67.36	0.759	99.44	99.26	99.12	98.69	0.323	90.17	<u>90.07</u>	88.73	83.26	0.444
FedNTD [169]	51.43	35.06	37.37	22.18	0.647	32.48	28.92	24.36	21.21	0.492	85.47	31.41	78.87	30.18	0.930	83.67	<u>79.23</u>	70.12	52.04	0.782

$\mathcal{E}$ is evaluated under the $\beta = 0.5$. Best in bold and second with underline. / denotes that these methods are not available for these metrics, i.e., the value is zero or NaN. These notes also apply to the other tables. Please refer to § 3.1 for metrics definition and § 6.1 for experimental analysis.

configuration to optimize towards a satisfying condition to avoid the potential severe overfitting problem.

2) *Robust Aggregation Defense*: A group of solutions addresses this issue during the aggregation stage by excluding the malicious signals such as weights and gradients, from suspicious clients through anomaly behavior detection and dynamic weight allocation [44], [130], [268], [281], [282]. Importantly, a part of these solutions are initially designed to defend against Byzantine attacks (as discussed in Section IV-B) but have also proven to be effective in countering backdoor attacks. For example, DimKrum [282] identifies abnormal and malicious client updates on a small fraction of dimensions with higher backdoor strengths, building on the Krum [130]. RLR [281] adjusts the server learning rate through the sign information of client updates in terms of the dimension and round aspects. However, these solutions typically assume that each client distribution obeys the independent and identically distributed property, which does not hold for the heterogeneous federated learning setting [26], [68], [95], [283].

3) *Certified Robustness Defense*: This family of methods aims to provide the certified robustness guarantee for each testing example, i.e., the prediction would not change even if some features in local training data of malicious clients have been modified within a certain constraint [280], [284], [285], [286], [287], [288], [289]. Specifically, ProvableFL [288] acquires multiple global models via optimizing on a randomly selected client subset. When inferring the testing example label, we take the majority vote results among the global models. CRFL [287] controls the global smoothness via the clipping and smoothing parameter operations. FLIP [289] combines trigger inversion techniques during the federated training process. Yet, it either requires the storage cost for a large number of models or brings performance degradation due to heavy parameter interpretation operation. Besides, some pre-define possible attack types and thus are not suitable for the open world.

D. Relation Between Byzantine and Backdoor Defense

In exploring the relationship between Byzantine faults and backdoor defense solutions, we begin with a preliminary review of the objectives behind these two types of attacks. Building upon the attack goals, Byzantine Attack primarily focuses on untargeted disruptions, aiming to hinder the global model

convergence or to degrade federated performance [44], [47], [49], [137]. Backdoor Attack seeks to compromise accuracy on specific test inputs while maintaining high accuracy on the original evaluation distribution [53], [54], [104], [135]. Consequently, the mitigation of such malicious impacts, a strategy known as Robust Aggregation Defense, has garnered considerable research interest as a means to defend against both Byzantine Attack and Backdoor Attack in heterogeneous federated learning. Broadly speaking, Robust Aggregation Defense involves the exclusion of suspect weights and gradients from ambiguous clients through anomaly detection [130], [140], [282], [290], [291], [292]. Additionally, with respect to Backdoor Attack, the concept of purifying the model from triggered patterns emerges as a novel defensive paradigm, which can be categorized into two approaches. Specifically, Model Refinement Defense aims to refine the aggregated model through techniques such as distillation [236], pruning [279], fine-tuning [277], and specification [280]. On the other hand, Certified Robustness Defense explores certified robustness against backdoor attacks, mainly by modifying client-specific optimization strategies, such as clipping and smoothing methods [287], leveraging the Hessian matrix [293], and employing majority voting [288]. Overall, both byzantine and backdoor defense strategies enhance robustness by eliminating malicious aggregation weight, with the latter also considering the federated model purification through parameter updates or certified robustness establishment.

E. Free Rider Detection

To deal with the free rider attacker, existing defense solutions could be basically divided into two categories.

1) *Outlier Parameter Detection*: The stream of solutions focuses on detecting the outlier of model parameters to mitigate the free rider effect [131], [294]. [131] proposes to require the benign clients to pre-train the autoencoder module and further detect free riders through the learned features of model update parameters. WEF-Defense [294] observes that free riders and benign clients appear the differences in the dynamic training progress and require clients to upload the weight evolving frequency to discriminate the free riders. To some extent, this group shares similar insights with the Byzantine Tolerance discussed in Section IV-B and the Robust Aggregation Defense strategy for Backdoor Defense outlined in Section IV-C2.

2) *Client Contribution Evaluation*: Considering that free riders do not contribute to the federated system, several methods have devoted themselves to designing reasonable contribution evaluation methods for the federation [133], [287], [295], [296], [297], [298]. For instance, RRFL [297] calculates the client reputation by examining their contributions based on the uploaded gradients. RepConFL [295] introduces the reputation to measure reliability and trustworthiness via multiweight subjective logic model. DPFL [296] designs a 3-D contract-based incentive mechanism to combine the data owners contribution with heterogeneous costs consisting of computation, communication, and privacy cost. Recently, BESIFL [298] further considers the fully decentralized federated system and develops an accuracy-based malicious node detection strategy. Additionally, it proposes a contribution-based incentive mechanism that utilizes a token-based reward scheme. This mechanism is designed to select credible clients that are most likely to contribute positively to a federated learning system. Notably, the Client Contribution Evaluation paradigm shares a certain similarity with Collaboration Fairness in Section V-A.

F. Fairness Metrics

Federated learning is not a *utopia*, which describes an imaginary community or society that holds highly desirable or near-perfect qualities for its members [299]. Federated learning unavoidably brings the interest conflict problem. Specifically, the client spends both computation and communication costs to optimize and upload the federated models [16], [300], [301], [302], [303]. Besides, with respect to the quality and quantity of the data, the data value varies among different clients [304], [305]. Their contributions to the federated system would not be suitable to regard equally. This federated learning requires incentive mechanisms to distinctly measure contribution and reward benefits for each client [57], [58]. Existing Fair Federated Learning strategies mainly deal with two types of unfairness: Reward Conflict and Prediction Biases, and thus naturally group into two methodology objective streams: Collaboration Fairness (Section V-A) and Performance Fairness (Section V-B). We provide the respective evaluation metrics as the following form.

- *Contribution Match Degree*. $\mathcal{E}$: Evaluating the client importance acts as an important step to reasonably allocate the profit [61], [306]. We take inspiration from the Shapley value [156] to measure the client marginal contribution. We argue that the reward mechanism is normally based on the pre-defined aggregation weight (α in (1)) to allocate reward. However, the realistic the client contribution could be could be directly reflected in the performance change. Specifically, we conduct the leave-one-out experiment and define the federated model without i client as $w^{-i} = \frac{w - \alpha_i w_i}{1 - \alpha_i}$. We measure the performance drop, Γ_i , if we remove the query client i . Then, we calculate $\mathcal{E}$ as:

$$\Gamma_i = \bar{\mathcal{A}} - \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \mathcal{A}_{-i}^u,$$

$$\Gamma = \left[\dots, \frac{\Gamma_i}{\sum_{i \in K} \Gamma_i}, \dots \right] \in \mathbb{R}^{|\mathcal{K}|},$$

$$\mathcal{E} = \frac{\Gamma \cdot \alpha}{\|\Gamma\|_2 \|\alpha\|_2}. \quad (13)$$

The (13) reveals that a Collaboration Fairness methodology would achieve a higher $\mathcal{E}$ to reasonably allocate interest.

- *Performance Deviation*. $\mathcal{V}$: Performance Fairness aims to maintain both **high** average accuracy and **uniform** accuracy across distribution. Following [63], [162], [307], we utilize the standard deviation to measure the performance inconsistency of the selected algorithm across different testing distributions.

$$\mathcal{V} = \sqrt{\frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} (\mathcal{A}^u - \bar{\mathcal{A}})^2}, \quad (14)$$

where $\bar{\mathcal{A}}$ denotes the average performance on the overall testing distributions. The larger the $\mathcal{V}$ is, the more biased the federated performance is. Therefore, federated performance fairness focuses on maintaining a **high** mean domain performance $\bar{\mathcal{A}}$ and a **low** performance variance $\mathcal{V}$.

V. FAIR FEDERATED LEARNING

A. Collaboration Fairness

Contribution evaluation plays a crucial role in federated learning by assessing each client's contribution in a privacy-friendly manner [56], [308]. Specifically, when one party contributes more, it may feel unfairly treated if all parties receive the same compensation regardless of their differing contributions. Hence, a fair contribution evaluation mechanism motivates clients to join the collaboration. Existing approaches basically follow the following ideologies.

- 1) *Individual Contribution Evaluation*: The individualized assessment paradigm focuses on measuring the client contribution based on individual contribution via the locally relevant information and the specific task performance. Specifically, the client-representative information normally builds on data collection cost [309], contract theory [310], [311], Stackelberg Game [312], [313], and computation bids [314]. As for the specific task performance, some methods are based on the reputation mechanism to record the clients contributions [61], [297], [310], [314], [315], [315], [316]. For example, CFFL [61] evaluates the fairness by evaluating the validation performance on each client to represent the reputation metric. RRFL [297] considers the updates divergence between each local and shared global model to represent the reputation value. Besides, several papers introduce different contribution metrics such as client bid and resource quality score [314] and pair-wise mutual evaluation mechanism [56]. However, Individual Contribution Evaluation often assumes that both servers and clients are trustworthy, which can yield unreliable client contribution results in a semi-honest setting. Moreover, it may be less effective in dealing with significant data heterogeneity because clients with vastly different data distributions may exhibit noticeably different performances compared to others, resulting in the low reputation scores.

- 2) *Marginal Contribution Evaluation*: As federated learning is a multi-party collaboration paradigm, some work evaluates the client marginal contribution to the federated performance. Notably, existing solutions normally take inspiration from

the cooperative game theory [155], [157], [158], especially shapley value (SV) [156] to measure the player value. Related solutions focus on efficient shapely value measurement under the federated scenarios [107], [306], [317], [318], [319]. For example, CGSV [306] introduces the cosine gradient Shapley value to approximate the SV between the client and global updates. FedCE [107] estimates client contribution in both gradient and data space for medical image segmentation on the local evaluation. However, these methods still face a heavy computation burden because the computational complexity of calculating SV is $\mathcal{O}(2^n)$. Furthermore, some require auxiliary dataset for performance validation, posing the additional data collection requirement.

B. Performance Fairness

The performance disparity means that the federated model presents a biased prediction preference, which appears in certain protected groups classification failure because of overfitting certain clients at the expense of other clients. Existing Performance Fairness methods mainly focus on achieving uniform performance from the local optimization and parameter aggregation aspects [62], [63], [164], [320].

1) *Performance Debias Optimization*: A set of solutions proposes to mitigate the biased global performance by modifying the local objective function to satisfy the specific fairness constraints. One type focuses on the client with worst performance [62], [64], [321]. For example, AFL [62] focuses on the single worst client and introduces the min-max approach to avoid overfitting to the particular client. However, naively focusing on the single worst objective can drag on the overall model utility [58]. Another group leverages the multi-objective optimization [106], [109], [307], [322], [323]. For instance, FedMGDA+ [307] designs a constraint multiple gradient descent for a common descent direction. FCFL [109] utilizes the surrogate maximum function to consider the multi-objective optimization to achieve Pareto optimality by controlling the gradient direction. However, these algorithms often assume that clients are benign and do not engage in malicious behavior against the federation. If a client maliciously reports its local empirical loss, it can lead to misleading updating directions [58] and brings the federated optimization failure.

2) *Performance Debias Reweighting*: A mainstream of solutions has been developed to reweight the parameter aggregation to alleviate biased performance. These methods primarily rely on various parameter variance signals, such as empirical losses [63], gradients [162], [324], predictive risk [107], [110], [266], [325]. In particular, qFFL [63] increases gradient weight of clients with lower losses by recalibrating the aggregate loss. FedHEAL [324] mitigates gradient conflict by aggregating gradient with consistent update directions and prevents distributions bias by minimizing the variance of update distances. FedCE [107] reduces the generalization gaps variance among different source domains to encourage optimization flatness and fairness. However, gradient estimation is based on the previous rounds and fails to synchronize with the latest updates. Additionally, prediction evaluation involves additional validation datasets, posing a serious data requirement.

VI. BENCHMARK

Preliminary, we provide the detailed experimental datasets and implementation details in the appendix. We tabulate the performance of previously discussed solutions. For each of the reviewed fields, the most widely used dataset is selected for performance benchmarking. The performance scores are gathered from the reproduced results. As different methods are with different code bases and levels of optimization, it is hard to make completely fair comparison. Besides, for a small set of methods whose implementations are not well organized or publicly available, we directly borrow them from the unofficial code repository.

A. Generalization Benchmark

1) *Evaluation Metrics*: Cross-Client Accuracy $\mathcal{A}^u$ is widely adopted in the Cross-Client Shift: Label Skew and Domain Skew settings. We further denote Out-Client Accuracy $\mathcal{A}^o$ under the Out-Client Shift for generalizable performance evaluation.

2) *Results*: As shown in Table V, Cifar-10 [326], Cifar-100 [326], MNIST [159], and Fashion-MNIST [327] are arguably popular datasets for the Label Skew setting. We summarize the results of ten methods on these four datasets. It reveals the first method FedAvg [89] proposed in 2017, to recent complicated solutions [169]. For further comparison, we visualize the training curves of the testing evaluation accuracy along with the training process under the $\beta=0.5$. Furthermore, as for the Domain Skew scenario, we adopt the widely federated scenarios, i.e., Digits [159], [160], [328], [329], Office Caltech [330], and [331]. SCAFFOLD [164] and FedProto [98] present relative competitive performance. Regarding the Out-Client Shift task, we take into account the specialized Federated Domain Adaptation Section III-C1 and Federated Domain Generalization Section III-C2 paradigms. Notably, Federated Domain Adaptation utilizes the unlabeled the unknown domain distribution during the training process and achieves the clear Out-Client Accuracyimprovement. For example, KD3A achieves the 67.16 accuracy on the Office Caltech scenario.

B. Robustness Benchmark

1) *Evaluation Metrics*: $\mathcal{A}_{Byz}^u$ means the testing accuracy under the Byzantine Attack. And thus, Accuracy Decline Impact $\mathcal{I}$ denotes the decreased accuracy, compared with the benign federation results. Similarly, Attack Success Rate $\mathcal{R}^u$ evals the performance on the backdoor polluted datasets.

2) *Results*: We conduct the comparison under four widely adopted datasets, i.e., Cifar-10, Fashion-MNIST, MNIST, and USPS datasets with both Data-Based Byzantine Attack: Pair Flipping and Symmetry Flipping, and Model-Based Byzantine Attack: Random Noise and Min-Sum. We select the popular Byzantine Tolerance methods from three types: Distance Base Tolerance, Statistics Distribution Tolerance, Proxy Dataset Tolerance. The results show that DnC reaches a relatively satisfying performance under different Byzantine Attack. Proxy Dataset Tolerance appears the obvious disadvantage, additional related proxy dataset. The Table VI evaluates the Backdoor Attack results under two popular forms: Bac, and Sem Bac. We further

TABLE VI
QUANTITATIVE BYZANTINE ATTACK RESULTS IN TERM OF $\mathcal{A}^u$, $\mathcal{A}_{Byz}^u$, AND $\mathcal{I}$ (11) ON CIFAR-10, MNIST, AND FASHION-MNIST SCENARIOS

Methods	Cifar-10						Fashion-MNIST						MNIST						USPS					
	$\beta = 0.5$			$\beta = 0.3$			$\beta = 0.5$			$\beta = 0.3$			$\beta = 0.5$			$\beta = 0.3$			$\beta = 0.5$			$\beta = 0.3$		
	$\Upsilon = 0.2$	$\Upsilon = 0.4$		$\Upsilon = 0.2$	$\Upsilon = 0.4$		$\Upsilon = 0.2$	$\Upsilon = 0.4$		$\Upsilon = 0.2$	$\Upsilon = 0.4$		$\Upsilon = 0.2$	$\Upsilon = 0.4$		$\Upsilon = 0.2$	$\Upsilon = 0.4$		$\Upsilon = 0.2$	$\Upsilon = 0.4$		$\Upsilon = 0.2$	$\Upsilon = 0.4$	
	$\mathcal{A}_{Byz}^u$	$\mathcal{A}_{Byz}^u$	$\mathcal{I}$	$\mathcal{A}_{Byz}^u$	$\mathcal{A}_{Byz}^u$	$\mathcal{I}$	$\mathcal{A}_{Byz}^u$	$\mathcal{A}_{Byz}^u$	$\mathcal{I}$	$\mathcal{A}_{Byz}^u$	$\mathcal{A}_{Byz}^u$	$\mathcal{I}$	$\mathcal{A}_{Byz}^u$	$\mathcal{A}_{Byz}^u$	$\mathcal{I}$	$\mathcal{A}_{Byz}^u$	$\mathcal{A}_{Byz}^u$	$\mathcal{I}$	$\mathcal{A}_{Byz}^u$	$\mathcal{A}_{Byz}^u$	$\mathcal{I}$	$\mathcal{A}_{Byz}^u$	$\mathcal{A}_{Byz}^u$	$\mathcal{I}$
FedProx [27]	$\mathcal{A}^u$:67.16			$\mathcal{A}^u$:64.88			$\mathcal{A}^u$:89.97			$\mathcal{A}^u$:88.69			$\mathcal{A}^u$:99.41			$\mathcal{A}^u$:99.32			$\mathcal{A}^u$:96.70			$\mathcal{A}^u$:96.69		
Pair Flipping																								
Multi Krum [130]	50.21	46.85	20.31	46.99	43.91	20.82	82.20	47.59	42.38	80.79	82.51	6.18	10.18	11.35	88.06	10.43	11.35	87.97	50.83	93.52	3.18	93.41	51.11	45.58
Bulyan [44]	46.88	44.06	20.68	10.00	10.00	54.88	82.62	80.76	9.21	78.00	73.57	15.12	97.01	98.18	1.23	93.21	92.13	7.19	93.21	92.13	4.57	86.04	87.20	9.49
Trim Median [268]	51.70	45.77	21.39	19.94	10.67	54.21	84.18	78.09	11.88	81.76	77.89	10.8	98.57	94.62	4.79	93.25	92.90	6.42	94.85	94.33	2.37	91.72	92.05	0.64
FoolsGold [140]	60.09	56.80	10.36	50.81	57.98	6.90	86.97	86.07	3.90	85.65	81.50	7.19	97.25	97.80	1.61	98.05	97.22	2.10	77.69	91.77	4.93	87.90	77.23	19.46
DnC [137]	62.67	58.38	8.78	60.41	59.96	4.92	87.54	87.76	2.21	87.22	88.24	0.45	99.33	99.07	0.34	98.85	98.70	0.62	95.94	95.16	1.54	95.07	95.08	1.61
FLTrust [270]	/	/	/	/	/	/	/	/	/	/	/	/	11.35	11.35	88.06	11.35	78.68	20.64	13.15	13.15	83.55	13.15	13.15	83.54
Sageflow [271]	/	/	/	/	/	/	/	/	/	/	/	/	99.28	99.03	0.38	99.02	98.73	0.59	95.36	94.34	2.36	96.15	95.37	1.32
RFA [269]	66.84	66.31	0.85	62.28	61.54	3.34	89.67	89.73	0.24	88.18	88.73	-0.04	99.12	99.10	0.31	98.97	98.91	0.41	96.12	95.56	1.14	96.30	96.08	0.61
Symmetry Flipping																								
Multi Krum [130]	52.18	46.48	20.68	49.03	50.56	14.32	81.87	85.52	4.45	82.14	81.76	6.93	10.02	91.76	7.65	11.35	92.72	6.60	81.20	93.06	3.64	84.12	93.79	2.90
Bulyan [44]	50.73	38.38	28.78	14.55	27.01	37.87	84.15	82.15	7.82	79.51	74.93	13.76	97.16	97.52	1.89	87.10	91.66	7.66	91.46	89.71	6.99	89.94	87.93	8.76
Trim Median [268]	53.24	49.82	17.34	34.46	39.24	25.64	84.61	84.39	5.58	80.49	81.48	7.21	98.50	98.08	1.33	92.16	96.25	3.07	93.46	92.23	4.47	93.32	93.70	2.99
FoolsGold [140]	61.37	59.34	7.82	58.35	54.97	9.91	69.15	86.30	3.67	82.34	84.27	4.42	98.46	97.77	1.64	95.90	90.45	8.87	83.02	78.07	18.63	75.72	73.92	22.77
DnC [137]	62.57	58.12	9.04	61.94	59.51	5.37	88.15	87.23	12.74	86.33	87.83	0.86	99.31	98.99	0.42	98.63	98.63	0.69	95.86	94.70	2.00	94.98	93.64	3.05
FLTrust [270]	/	/	/	/	/	/	/	/	/	/	/	/	11.35	70.09	29.32	11.35	67.29	32.03	60.41	52.83	43.87	59.31	13.15	83.54
Sageflow [271]	/	/	/	/	/	/	/	/	/	/	/	/	98.86	98.75	0.66	98.51	98.31	1.01	94.08	92.32	4.38	95.33	92.93	3.76
RFA [269]	63.43	61.67	5.49	62.78	60.13	4.75	89.44	88.30	11.67	87.73	87.49	1.20	99.00	99.06	0.35	98.78	98.65	0.67	95.80	94.57	2.13	95.98	95.47	1.22
Random Noise																								
Multi Krum [130]	10.00	13.06	54.1	29.25	14.11	50.77	10.00	21.71	68.26	75.55	25.60	63.09	11.35	13.42	85.99	11.35	21.04	78.28	89.25	15.07	81.63	13.15	26.79	69.90
Bulyan [44]	51.04	51.34	15.82	42.09	49.29	15.59	82.70	87.24	2.73	81.70	86.43	2.26	98.74	98.63	0.78	91.95	98.32	1.00	94.27	94.51	2.19	92.59	95.34	1.35
Trim Median [268]	53.87	51.92	15.24	50.24	50.21	14.67	85.94	85.66	4.31	82.32	85.61	3.08	98.86	98.85	0.56	94.36	98.18	1.14	94.80	13.15	83.55	95.66	95.59	1.10
FoolsGold [140]	50.01	32.85	34.31	49.60	27.45	37.43	85.98	35.82	54.15	76.86	83.58	5.11	98.46	37.62	61.79	87.91	78.90	20.42	85.36	22.55	74.15	54.10	55.92	40.77
DnC [137]	59.64	56.95	10.21	60.00	56.45	8.43	87.81	87.72	2.25	87.26	87.66	1.03	99.31	98.97	0.44	98.78	98.85	0.47	95.73	94.60	2.10	95.31	94.28	2.41
FLTrust [270]	/	/	/	/	/	/	/	/	/	/	/	/	11.35	11.35	88.06	11.35	11.35	87.97	36.53	13.15	83.55	13.15	13.15	83.54
Sageflow [271]	/	/	/	/	/	/	/	/	/	/	/	/	98.76	96.75	2.66	93.14	89.85	9.47	92.40	78.20	18.50	86.02	75.63	21.06
RFA [269]	56.37	10.64	56.52	55.88	15.45	49.43	87.11	64.10	25.87	85.32	72.30	16.39	99.15	95.40	4.01	98.26	94.01	5.31	94.67	67.49	29.21	95.35	53.08	43.61
Min-Sum																								
Multi Krum [130]	10.00	10.90	56.26	42.20	10.02	54.86	10.00	11.02	78.95	80.78	10.00	78.69	11.35	23.17	76.24	10.43	11.35	87.97	13.15	15.96	80.74	13.15	13.15	83.54
Bulyan [44]	51.49	51.00	16.16	42.99	40.07	24.81	84.64	85.84	4.13	80.23	84.21	4.48	98.60	94.38	5.03	92.40	90.14	9.18	94.88	85.91	10.79	92.91	93.36	3.33
Trim Median [268]	53.62	53.71	13.45	49.58	51.76	13.12	84.64	85.71	4.26	83.24	85.41	3.28	98.77	98.76	0.65	96.80	92.90	6.42	95.12	95.75	0.95	94.22	95.45	1.24
FoolsGold [140]	52.26	10.00	57.16	47.83	10.00	54.88	80.58	14.80	75.17	80.20	19.36	69.33	97.18	16.87	82.54	98.71	97.22	2.10	69.49	15.04	81.66	64.16	13.12	83.57
DnC [137]	61.11	55.52	11.84	60.29	55.83	9.05	87.63	87.80	2.17	87.25	88.01	0.68	99.19	99.20	0.21	98.80	98.70	0.62	95.34	94.51	2.19	94.93	95.35	1.34
FLTrust [270]	/	/	/	/	/	/	/	/	/	/	/	/	61.57	12.99	86.42	11.35	11.35	87.97	13.15	15.04	81.66	13.15	14.09	82.60
Sageflow [271]	/	/	/	/	/	/	/	/	/	/	/	/	98.59	92.85	6.56	92.30	85.01	14.31	87.07	14.09	82.61	81.95	50.59	46.1
RFA [269]	51.90	11.40	55.76	60.29	14.22	50.66	87.40	22.83	67.14	85.71	61.18	27.51	99.05	94.39	5.02	98.80	98.91	0.41	94.65	71.23	25.47	94.93	57.83	38.86

FLTrust and sageflow utilizes SVHN as the proxy. The local optimization is FedProx [27] with $\mu = 0.01$. See Byzantine Tolerance comparison in § 6.2.

evaluate two Backdoor Defense: RLR [281] and CRFL [287]. It shows that RFA and CRFL achieves satisfying defensive performance.

C. Fairness Benchmark

1) *Evaluation Metrics*: Contribution Match Degree: $\mathcal{E}$ and Performance Deviation $\mathcal{V}$ respectively evaluates the Collaboration Fairness and Performance Fairness.

2) *Results*: Few of the existing federated optimization takes the Collaboration Fairness into federated objective account. Besides, Collaboration Fairness also is largely impended under large local data distribution diversity such as the Domain Skew. Regarding the Performance Fairness, existing methods basically focus on minimizing the weighted empirical loss and thus bring the imbalanced performance. Notably, global network utilization and server adaptive optimization seem to roundly alleviate the imbalanced performance on the multiple domains.

VII. OUTLOOK

A. Future Direction

Based on the reviewed research, we list several future research directions that we believe should be pursued.

- *Generalization and Robustness Dilemma*: Distributed data inherently exhibits the heterogeneous property and generalization aims to incorporate diverse client knowledge. Concurrently, Robustness aims to identify and mitigate the malicious clients

effect on the federated system [332], [333]. However, data heterogeneity in federated learning brings that certain local distributions deviate significantly from the primary trend and may be erroneously categorized as malicious ones. This misclassification restricts the effectiveness of federated performance because some benign clients are rejected from the federation. We further provide other research direction discussion in the Appendix.

B. Conclusion

To our knowledge, this is the first survey to comprehensively review recent progress in federated learning from generalization, robustness, and fairness aspects. We provided the reader with the necessary background knowledge and summarized more than 100 federated methods according to various criteria, including task settings, learning strategies, and technique contributions. We also present benchmarking results of 8 widely used federated datasets. We discuss the results and provide insight into the shape of future research directions and open problems in the field. In conclusion, federated learning has achieved notable progress attributed to the striking research development, but several challenges still lie ahead.

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Wenke Huang received the BE degree from the School of Computer Science, Wuhan University, Wuhan, China, in 2021. He is currently working toward the PhD degree with the School of Computer Science, Wuhan University. His research interests focus on Federated Learning.



Mang Ye (Senior Member, IEEE) received the PhD degree in computer science from Hong Kong Baptist University, in 2019. He is currently a full professor with the School of Computer Science, Wuhan University, Wuhan, China. He has published more than 100 articles in top-tier venues. He serves as the associate editor for the Journal of Electronic Imaging and CAAI Transactions on Intelligence Technology. His research interests focus on computer vision, pattern recognition, and federated learning.



Zekun Shi received the BE degree from Wuhan University, China, in 2021. He is currently a researcher in Xiaomi Corporation, Beijing, China. His research interests include computer vision and federated learning.



Guancheng Wan is currently working toward the BE degree with the School of Computer Science, Wuhan University, Wuhan, China. His research interests are graph machine learning and federated learning.



He Li received the master's degree in computer science from Trinity College Dublin, Ireland, 2021. He is currently working toward the PhD degree with the School of Computer Science, Wuhan University, China. His research interests focus on Person Re-Identification and AI for Medicine.



Bo Du (Senior Member, IEEE) received the PhD degree in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China, in 2010. He is a professor with the School of Computer Science, Wuhan University. He has more than 60 research articles published in the IEEE TGRS, TIP, JSTARS, and GRSL. Thirteen of them are ESI hot articles or highly cited articles. His major research interests include pattern recognition, hyperspectral image processing, machine learning, and signal processing. He was a recipient of the Distinguished Paper Award from IJCAI 2018, the Best Paper Award of the IEEE Whispers 2018, the Champion Award of the IEEE Data Fusion Contest 2018, the Best Reviewer Award from the IEEE GRSS for his service to the IEEE Journal of Selected Topics in Earth Observations and Applied Remote Sensing, in 2011, and the ACM rising star awards for his academic progress, in 2015. He was the session chair of the IGARSS 2018/2016 and the 4th IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing.



Qiang Yang (Fellow, IEEE) received the graduation degree from Peking University and PhD degree from the Department of Computer Science, the University of Maryland, College Park, in 1989 and had been a faculty member, the University of Waterloo between 1989 and 1995. He is a fellow of Canadian Academy of Engineering (CAE) and Royal Society of Canada (RSC), Chief Artificial Intelligence Officer of WeBank, a chair professor of Computer Science and Engineering Department at Hong Kong University of Science and Technology (HKUST). He is the Conference Chair of AAAI-21, the Honorary vice president of Chinese Association for Artificial Intelligence(CAAI), the president of Hong Kong Society of Artificial Intelligence and Robotics(HKSAIR), the President of Investment Technology League (ITL) and Open Islands Privacy-Computing Open source Community. He is a fellow of AAAI, ACM, CAAI, IAPR, and AAAS. He was the Founding Editor in Chief of the ACM Transactions on Intelligent Systems and Technology (ACM TIST) and the Founding Editor in Chief of IEEE Transactions on Big Data (IEEE TBD). He received the ACM SIGKDD Distinguished Service Award, in 2017 and the Wu Wenjun outstanding contribution award of artificial intelligence, in 2019. His research interests are artificial intelligence, machine learning, data mining, and planning. He had been the Founding Director of the Huawei's Noah's Ark Research Lab between 2012 and 2015, the Founding Director of HKUST's Big Data Institute, the Founder of 4Paradigm and the President of IJCAI (2017-2019). His latest books are Transfer Learning, Federated Learning, Privacy-preserving Computing, and Practicing Federated Learning.